

mini-OEM Mesh Rider Radio - 2400~2482 MHz (Wi-Fi band)



RM-2450-12M3 ([2023 Update](#)) / RM-2450-12M4 ([SOC Variant](#))

2023 Update

The Doodle Labs mini and nano -OEM Mesh Rider Radios were updated in Q1 of 2023. Customers are encouraged to switch over to the newer design. Legacy hardware should not be used for new designs. Look for updated ordering codes in the model's datasheet.



Change List

- 1. Primary connector replaced by 6 connectors with different hardware interfaces
- 2. Added USB Host connector
- 3. Added 3x GPIO

SOC Variant

The SOC variant (nano-OEM / mini-OEM) includes 2 x 18-pin headers, enabling direct PCB mount for enhanced hardware connectivity. This design provides additional flexibility for integrating the radios into customizable embedded applications where space and connectivity are mission critical.



Change List

- 1. This variant comes in 2 x 18-pin headers for a direct PCB mount option.
- 2. New SOC variant Evaluation Kits (EVK) for integration support.

RM-2450-12M3 ([2023 Update](#)) / RM-2450-12M4 ([SOC Variant](#))

Datasheet	-
mini-OEM Mesh Rider Radio - 2400~2482 MHz (Wi-Fi band)	
Getting Started w/ the mini-OEM	-

General Hardware Integration Guides:

[Hardware Integration Guidelines](#)
[Mesh Rider Radio Hardware Interfaces](#)

Model-Specific Hardware Integration Guides:

[mini-OEM \(2023 Update\)](#)

General Software Setup Guides:

[Getting Started](#)
[Flashing new firmware](#)

Hardware Information (2023 Update)



Radio Integration Information

2023 Update:

[Pin Diagrams](#)
[2D Drawings](#)
[3D Drawings](#)
[Heat Map](#)
[Cable Drawings](#)

Evaluation Kit Information

2023 Update:

[EVK Pin Diagram and Wiring Guide](#)
[EVK Schematic](#)
[EVK Layout](#)
[EVK 2D Drawing](#)
[EVK 3D Drawing](#)

Hardware Information (SOC Variant)



Radio Integration Information

SOC Variant:

- Pin Diagrams
- 2D Drawings
- 3D Drawings
- Heat Map
- Screws (Mechanical Drawing)

SOC Variant (Baseboard):

- 2D Drawings
- 3D Drawings

Evaluation Kit Information

SOC Variant:

- EVK Schematic
- EVK Layout
- EVK Pin Diagrams
- EVK 2D Drawing
- EVK 3D Drawing

Software Information



Mesh Rider Firmware Downloads

Latest Firmware

[June 2025 Release](#)

Older Firmware

[Older firmware \(not recommended\)](#)

Long-Term-Stable Firmware

[May 2023 Long-Term-Stable Release](#)

Antennas + Accessories



Accessories Information

Heat Sink

Antenna Information

Evaluation Kit Antenna Datasheet (2024)

Regulatory Information



FCC (United States)

IC (Canada)

CE (Europe)